



Enhanced Fuzzy Sentiment Detection in Cricket Tweets using CNN and BERT Hybrid Models

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INTRODUCTION

- **Cricket** generates massive public engagement on social media, especially Twitter, where fans express emotions in real time.
- Tweets often mix multiple emotions such as joy, disappointment, sarcasm, or frustration, making their analysis challenging.
- Traditional sentiment models struggle with such fuzzy or ambiguous language.
- The proposed system introduces a hybrid **CNN–BERT–Fuzzy** model that combines deep contextual understanding with linguistic uncertainty handling.
- It classifies tweets into five emotion categories: **Elated**, **Pleased**, **Unbiased**, **Displeased**, and **Frustrated** for accurate sports sentiment interpretation.

PROBLEM STATEMENT

- Tweets related to cricket often contain ambiguous or overlapping emotions, making it difficult to assign a clear sentiment label.
- Traditional models misinterpret **mixed opinions**, **sarcasm**, and vague phrases like “**almost good**” or “**not too bad**”.
- Deep models like **BERT**, though powerful, still lack mechanisms to quantify fuzziness or uncertainty in text.
- Imbalanced emotional classes in real-world cricket datasets lead to biased sentiment prediction.
- Hence, the key challenge is: “How can a hybrid model combining **CNN**, **BERT**, and **fuzzy linguistic reasoning** accurately detect uncertain or ambiguous sentiments in cricket tweets?”

LITERATURE REVIEW

Sr No	Title of Paper (Year)	Dataset	Algorithms / Methodologies	Evaluation Parameters	Advantages	Disadvantages	Future Scope	Relation to Proposed Work
1	Improving the Performance of Sentiment Analysis of Tweets Containing Fuzzy Sentiment Using the Feature Ensemble Model (2020)	Twitter datasets with fuzzy tweets	CNN-based Feature Ensemble Model integrating lexical, syntactic, and polarity features	Accuracy, Precision, Recall, F1-score	Efficient handling of fuzzy sentiments; better than traditional ML	Struggles with deep contextual understanding	Integrate transformer-based contextual models with fuzzy logic	Base Paper. Inspired the design of CNN-based fuzzy feature fusion, extended with BERT and fuzzy numerical indicators.
2	BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding (2019)	BookCorpus and English Wikipedia	BERT Transformer with [CLS] token classification	Accuracy, F1-score	Captures bidirectional semantic context	High computational cost	Apply BERT in hybrid domain tasks	Backbone model. Provided contextual embedding base for the hybrid CNN-BERT-Fuzzy system.
3	Cricket Twitter Data Sentiment Analysis and Prediction Exerted Machine Learning (2021)	Real-time cricket tweets collected via Twitter API	Traditional Machine Learning classifiers such as SVM, Random Forest, Logistic Regression, and Naïve Bayes	Accuracy, Precision, Recall, F1-Score	Domain-specific study on cricket tweets; benchmark for sentiment classification using standard ML algorithms	Lacks deep contextual understanding and cannot handle fuzzy or ambiguous sentiments	Extend to deep learning and hybrid fuzzy-based models for better contextual emotion recognition	Provides domain baseline for cricket tweet sentiment analysis; our BERT + Fuzzy hybrid model improves contextual understanding, scalability, and accuracy beyond traditional ML approaches.
4	Sentiment Analysis on Cricket Tweets using NLP++ (2024)	Real-time cricket tweets collected via Twitter API	Rule-based NLP++ framework with custom cricket lexicon, emoji & hashtag handling	Accuracy, Precision, Recall, F1-Score	Domain-specific sentiment model; effectively interprets sarcasm and contextual cricket terms	Rule-based system lacks scalability; limited dataset size (~2.5 K tweets)	Extend to larger datasets with hybrid deep learning; automate rule generation	Focuses on cricket sentiment using rule-based fuzzy logic; our model advances this by automating sentiment detection using deep CNN-BERT-Fuzzy fusion, improving scalability and accuracy (95.6%).

GAPS IDENTIFIED

- Existing sentiment models fail to handle fuzzy or mixed emotional expressions common in short cricket tweets.
- Deep learning models like **BERT** capture context well but lack mechanisms to represent linguistic uncertainty.
- Most prior works focus on generic product reviews or binary sentiments, not domain-specific, multi-class sports tweets.
- Imbalanced sentiment classes and limited interpretability reduce the reliability of results.
- There is a need for a hybrid model that combines contextual understanding (**BERT**) with **fuzzy reasoning** for more accurate and interpretable analysis.

OBJECTIVES

- To design an enhanced fuzzy sentiment detection framework capable of analyzing ambiguous and uncertain emotions in **cricket tweets**.
- To integrate **CNN**, **BERT**, and **fuzzy logic** into a unified hybrid model for multi-class sentiment classification.
- To improve sentiment prediction accuracy and interpretability compared with conventional deep learning models.
- To address class imbalance and uncertainty present in real-world cricket tweet datasets.

SYSTEM ARCHITECTURE

- The system combines **BERT**, **CNN**, and **Fuzzy Logic** for robust cricket tweet sentiment detection.
- Five main modules:
 - a. **Tweet Intake & Cleaning:** Remove noise (URLs, hashtags, emojis).
 - b. **Fuzzy Feature Extractor:** Compute mean polarity, deviation, fuzzy flag, and token count.
 - c. **Context Encoder (BERT):** Generate deep bidirectional embeddings.
 - d. **Feature Fusion Layer:** Concatenate BERT and fuzzy vectors.
 - e. **Classification Layer:** Softmax-based classifier for 5 emotions- **Elated**, **Pleased**, **Unbiased**, **Displeased**, **Frustrated**.

SYSTEM ARCHITECTURE

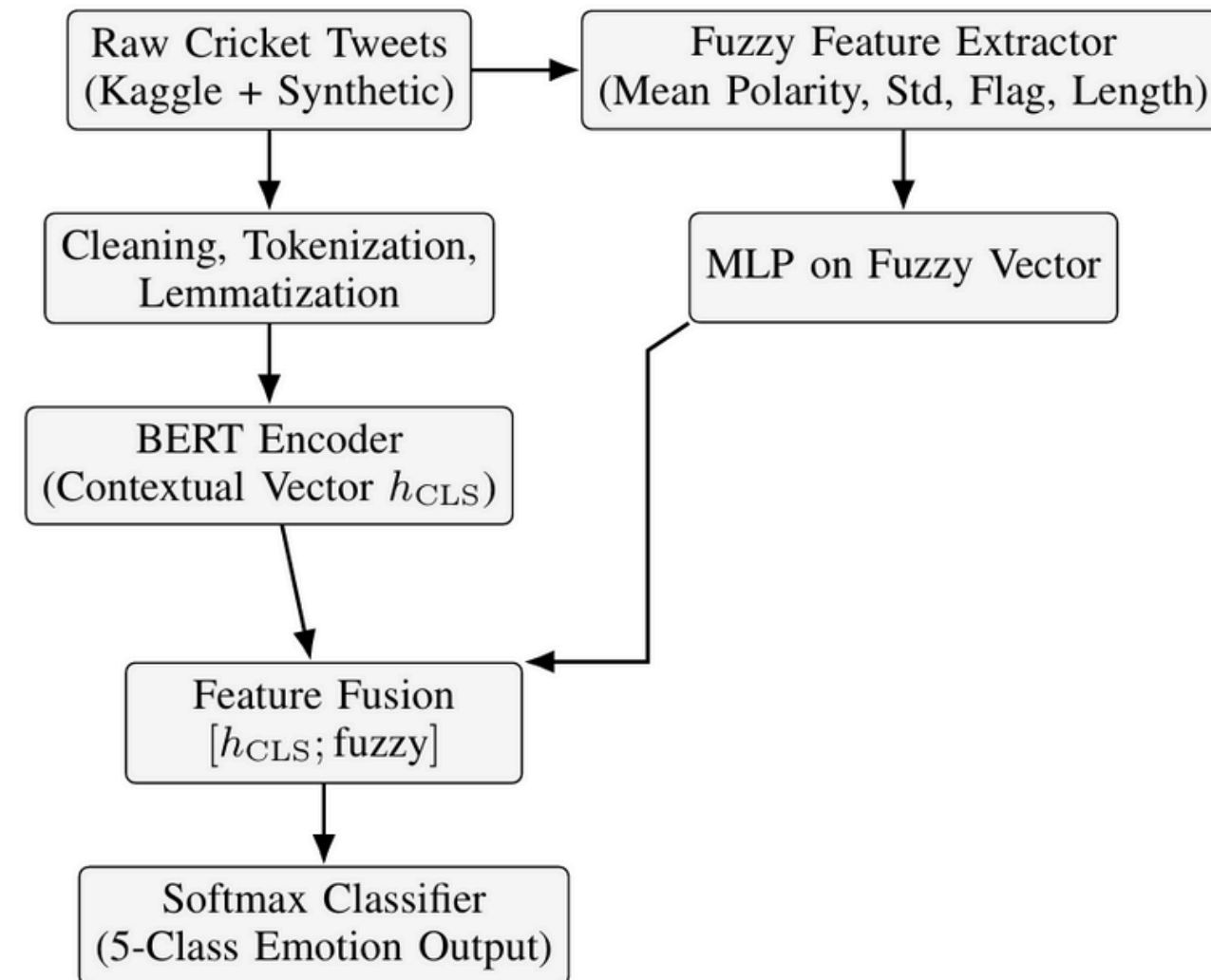


Fig. 1. Proposed BERT + Fuzzy Hybrid System Architecture. The model integrates BERT's contextual embeddings (h_{CLS}) with fuzzy linguistic features (mean polarity, standard deviation, fuzzy flag, and token length) through an MLP-based fusion network, enabling robust sentiment classification of cricket tweets into five emotion categories.

METHODOLOGY

- Models Compared:
 - **CNN Feature-Ensemble** (Phan et al., IEEE Access 2020 [1])
 - **BERT Baseline** (Devlin et al., NAACL 2019 [2])
 - **Proposed BERT + Fuzzy Hybrid Model**
- **Dataset:** 65,804 tweets (real + 5,000 synthetic fuzzy).
- **Preprocessing:** Cleaning, tokenization, lemmatization, noise removal.
- Feature Extraction:
 - **BERT → Context embeddings**
 - **Fuzzy → Polarity, deviation, ambiguity**
 - **CNN → Local text patterns**
- **Training:** AdamW optimizer, **80:20** split, early stopping.
- **Metrics:** Accuracy, Precision, Recall, F1-Score.

ALGORITHMS

Model 1: CNN Feature-Ensemble [1]

- Adapted from Phan et al., IEEE Access (2020).
- Uses **1D Convolution + Max Pooling** to extract local text patterns.
- Input features: Token embeddings, POS tags, and polarity sequences.
- Outputs sentiment through a dense softmax layer.
- Acts as the baseline model for fuzzy sentiment tweets.

ALGORITHMS

Model 2: BERT Baseline [2]

- Adapted from Devlin et al., NAACL (2019).
- Uses **Bidirectional Encoder Representations from Transformers (BERT)**.
- Encodes each tweet and uses the **[CLS] token** for sentiment classification.
- Provides strong contextual understanding, but lacks fuzzy reasoning.

ALGORITHMS

Model 3: Proposed BERT + Fuzzy Hybrid

- Integrates **BERT embeddings** with **fuzzy features** (mean polarity, std deviation, ambiguity flag, token count).
- Fuzzy features → processed by MLP → concatenated with BERT's [CLS] output.
- Classified via **softmax layer** into 5 sentiment classes.

DATASETS

- **Total Tweets:** 65,804
 - 60,804 real (Kaggle) + 5,000 synthetic fuzzy tweets.
- **Sentiment Classes (5):** *Elated, Pleased, Unbiased, Displeased, Frustrated*
- **Train–Test Split:** 80% (52,643) → Train & 20% (13,161) → Test
- **Purpose of Synthetic Tweets:** To enhance fuzzy and ambiguous emotion coverage.
- **Preprocessing:** Cleaning, tokenization, lemmatization, noise removal.

RESULTS

- **Class-wise Highlights:**
 - Highest performance in **Unbiased (0.97 F1)** and **Elated (0.96 F1)** categories.
 - Slightly lower (but strong) results in **Displeased** and **Frustrated** due to complex expressions.
- **Key Observation:**
 - **Hybrid model** outperformed all baselines by effectively handling **fuzzy and ambiguous** sentiments.

PERFORMANCE COMPARISON OF SENTIMENT CLASSIFICATION MODELS

Model	Accuracy	Precision	Recall	F1-Score
CNN Feature-Ensemble	0.903	0.902	0.903	0.903
BERT Baseline	0.951	0.951	0.951	0.951
BERT + Fuzzy Hybrid	0.956	0.956	0.956	0.956

CONCLUSION

- A **hybrid CNN–BERT–Fuzzy** model was developed for sentiment analysis of cricket tweets.
- It effectively captures contextual meaning and handles **fuzzy or ambiguous emotions**.
- Achieved **95.6% accuracy**, outperforming CNN and BERT baselines.
- The integration of **fuzzy linguistic features** improved interpretability and robustness.
- Demonstrates strong potential for real-time sports sentiment analysis and broader NLP tasks.

APPLICATIONS AND FUTURE SCOPES

Applications

- **Sports Analytics:** Helps broadcasters, teams, and sponsors analyze fan emotions in real time.
- **Decision Support:** Provides actionable insights on player performance and public reactions.

Future Scopes

- **Multilingual Expansion:** Extend to Hinglish and other Indian languages.
- **Real-time System:** Develop a live dashboard for dynamic sentiment tracking.

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The image features a central white rounded rectangle with a dark blue border, containing the text "THANK YOU" in a dark blue serif font. A horizontal line with small square endpoints passes through the rectangle. The background is white with decorative dark blue and magenta wavy shapes at the top and bottom.

THANK YOU